# Thermodynamics and Statistical mechanics:

Heat → Caused by temperature difference

1st law of thermodynamics: $\Delta U = \Delta Q + W$ (W: work done on it; Q: heat added)

$Q < 0 \Rightarrow$ system is cooled

$W < 0 \Rightarrow$ work is done by the system

2nd law of thermodynamics: If a system is at low temperature, spontaneously it cannot attain a higher temperature

Microstate, macrostate

↳ state of each individual particle must be specified

multiplicity ($\Omega$) = number of microstates corresponding to a given macrostate

Solid is modelled as a collection of oscillators with evenly spaced energy levels. To distribute $q$ units of energy among $N$ oscillators $\Rightarrow \Omega(N, q) = {}^{q+N-1}C_{N-1}$

Total multiplicity of any macrostate, $\Omega_{total}$ is the product of individual multiplicities

In an isolated system in thermal equilibrium, all accessible microstates are equally likely. But

and accessible microstates are equally likely. But some macrostates are more probable than others. (most likely macrostate is $10^{33}$ more probable than the least likely macrostate)

for large Einstein solid:

$$\Omega(N,q) = \frac{(q+N-1)!}{q!\,N!} = \frac{(q+N)!}{q!\,N!} \quad (q>>N)$$

$$N! = \sqrt{2\pi N}(N/e)^N \Rightarrow \ln(N!) = N\ln N - N + \frac{1}{2}\ln(2\pi N)$$

$$\Rightarrow \ln\Omega = (q+N)\ln(q+N) - (q+N) - q\ln q + q - N\ln N + N$$

$$(q+N)\ln\left(1+\frac{N}{q}\right) + (q+N)\ln q - q\ln q - N\ln N$$

$$= (q+N)N/q + N\ln(q/N) = N + N\ln(q/N) + N^2/q$$

$$\Rightarrow \Omega(N,q) = e^N \cdot (q/N)^N = (eq/N)^N \quad (q>>N)$$

Entropy $(S) = k\ln\Omega$ ; Always tends to increase (Any large system in equilibrium will be found in the macrostate with greatest entropy)

$$\Delta S = \Delta Q/T \quad (2^{nd} \text{ law of T.D})$$

Entropy of an ideal gas depends on energy, volume, no of particles. (↑ any of these ↑ the entropy)

Two weakly coupled systems (A and B) are allowed to exchange quantities and reaches equilibrium where total entropy is maximum $(S_A + S_B)$ with a constraint that total exchanged quantity is fixed.

$dS = \left(\frac{\partial S}{\partial U}\right)_{N,V} dU + \left(\frac{\partial S}{\partial V}\right)_{N,U} dV + \left(\frac{\partial S}{\partial N}\right)_{U,N} dN$

$dU = TdS - PdV + \mu dN$

Gibbs free Energy: $G = H - TS = U + PV - TS$

chemical work ↑

$dG = dU + PdV + VdP - TdS - SdT = -SdT + VdP + \mu dN$

mechanical work

$\mu = \left(\frac{dG}{dN}\right)_{T,P}$ = Gibbs free energy per particle

$S = -(dG/dT)_{P,N}$ ; $V = (dG/dP)_{T,N}$

$\mu$ is the Gibbs free energy per particle

particles tend to flow from higher chemical potential to lower chemical potential.

$\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings}$ (spontaneous if $\Delta S_{total} > 0$)

Gibbs Free Energy $dG = dH - TdS$ ; $dH = dU + PdV$

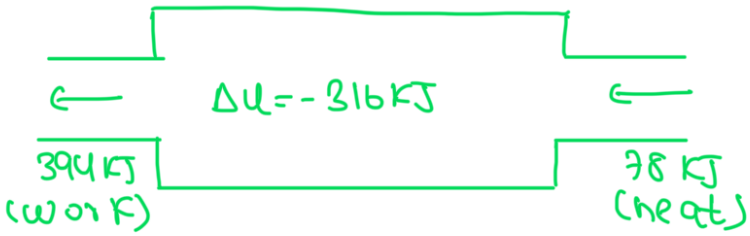
$\Rightarrow dG = dU + PdV - TdS = dw + dq + PdV - TdS$

for maximum work; it must occur reversibly

$\Rightarrow dG = dW_{rev} + PdV$ (at const temp & pressure)

$= dW_{rev,\ non\ expansion}$

$\Delta G \rightarrow$ maximum non expansion work that the system can do at constant T and P.




in soln: $2SO_4^{2-} + 2H^{\oplus} \longrightarrow 2HSO_4^{\ominus}$

-ve: $Pb + HSO_4^{\ominus} \longrightarrow PbSO_4 + H^+ + 2e$

+ve: $PbO_2 + HSO_4^{\ominus} + 3H^{\oplus} + 2e^{\ominus} \longrightarrow PbSO_4 + 2H_2O$

electrical work produced per $e^{\ominus}$

$$\frac{394\ kJ}{2 \times 6.02 \times 10^{23}} = 3.27 \times 10^{-19} = 2.04\ eV$$

## Canonical ensemble:

System in contact with a large reservoir

$U_R + E = E^0$ $\quad U_R \rightarrow$ Reservoir energy ; system energy $= E$

At equilibrium: $\Omega_S(E) = \Omega_R(U_R) = \Omega(E^0 - E)$

$$\Rightarrow \frac{d(\ln \Omega)}{d\mu} = \boxed{\frac{1}{kT} = \beta} \Rightarrow \Omega_R(E^0 - E) = \Omega_R(E^0) e^{-\beta E}$$

$$\Omega_S(E) = \Omega_R(E^0) e^{-\beta E} \qquad P(E) = \frac{1}{Z} e^{-\beta E}$$

($Z \rightarrow$ found by normalization)

$e^{-\beta E} \rightarrow$ Boltzmann factor (Boltzmann distribution)

Probability of finding the system in state $s$

$$\boxed{P(s) = \frac{1}{Z} e^{-E(s)/kT}} \qquad \boxed{Z = \sum e^{-E(s)/kT}}$$

$Z \rightarrow$ partition function

Average energy: $\bar{E} = \frac{\sum_s E(s)\, N(s)}{N} = \sum E(s) \frac{N(s)}{N}$

$$= \sum E(s) P(s) = \frac{1}{Z} \sum E(s) e^{-E(s)/kT}$$

## Grand canonical ensemble:

A system in thermal and diffusive contact with a large reservoir

Gibbs factor $= \boxed{e^{-[E(s) - \mu N(s)]/kT}}$

Grand partition function $Z = \sum_s e^{-[E(s) - \mu N(s)]/kT}$

$$P(s) = \frac{1}{\quad} e^{-[E(s) - \mu N(s)]/kT}$$

## classical and Quantum particles:

↓ distinguishable ↓ indistinguishable

If two particles are indistinguishable, the wave function must satisfy requirements depending on whether they are Fermions or Bosons.

### classical particles following Max-Boltz statistics

$$W_B = (N!) \prod_{i=0}^{\infty} \frac{(g_i)^{N_i}}{(N_i)!}$$

$W_B$ ↓ thermodynamic probability

$(g_i)^{N_i}$ → ways particles can be arranged within assigned energy levels

$(N_i)!$ → no of ways u can choose which of the N total distinguishable particles go into each energy level

- distinguishable
- → any no of particles in one state

### Quantum particles following Bose Einstein statistics

- → indistinguishable
- → any no of particles in one state

$$\Omega = \prod_i \frac{(g_i + n_i - 1)!}{n_i!\,(g_i - 1)!}$$

### Quantum particles following Fermi-Dirac statistics

- → indistinguishable
- → only one particle in one state

$$\Omega = \prod_i \frac{g_i!}{(g_i - n_i)!\,n_i!}$$

### distribution function of quantum gas

probability of state being occupied by n-Particles → $P(n) = 1\, e^{-n(\epsilon - \mu)\beta}$ ; $\epsilon$ → energy of

$\frac{1}{Z}$

state when occupied by a single particle

If particles are fermions, $n = 0$ or $1$

$$Z = 1 + e^{-(\epsilon-\mu)\beta}$$

Occupancy: $\sum n P(n) = \frac{1}{e^{(\epsilon-\mu)\beta}+1}$ → fermi-dirac distribution

$M$ @ $T = 0$ → Fermi Energy

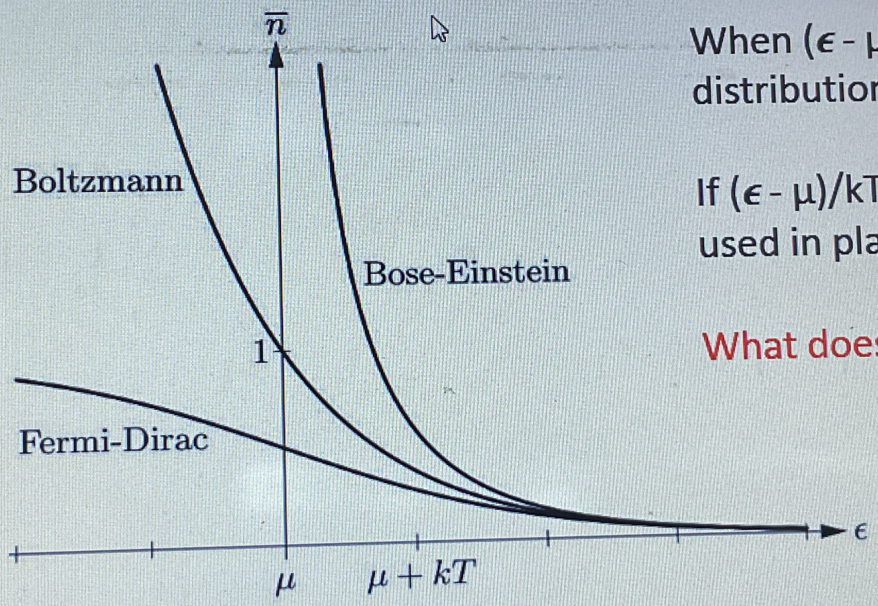
If particles are bosons, $n$ can be non-ve $Z$:

$$Z = \frac{1}{1 - e^{-(\epsilon-\mu)\beta}}$$

$\Rightarrow \bar{n} = \frac{1}{e^{(\epsilon-\mu)\beta} - 1}$ → Bose-Einstein distribution

($\mu < \epsilon$ always)

# Approximating as Maxwell-Boltzmann Distribution




When $(\epsilon - \mu)/kT >> 1$, the thr distributions become equal

If $(\epsilon - \mu)/kT > 3$, MB can be used in place of others

What does it mean $\overline{n} > 1$ ?

Approximating as Maxwell-Boltzmann

search
31°C Mostly cloudy
DELL

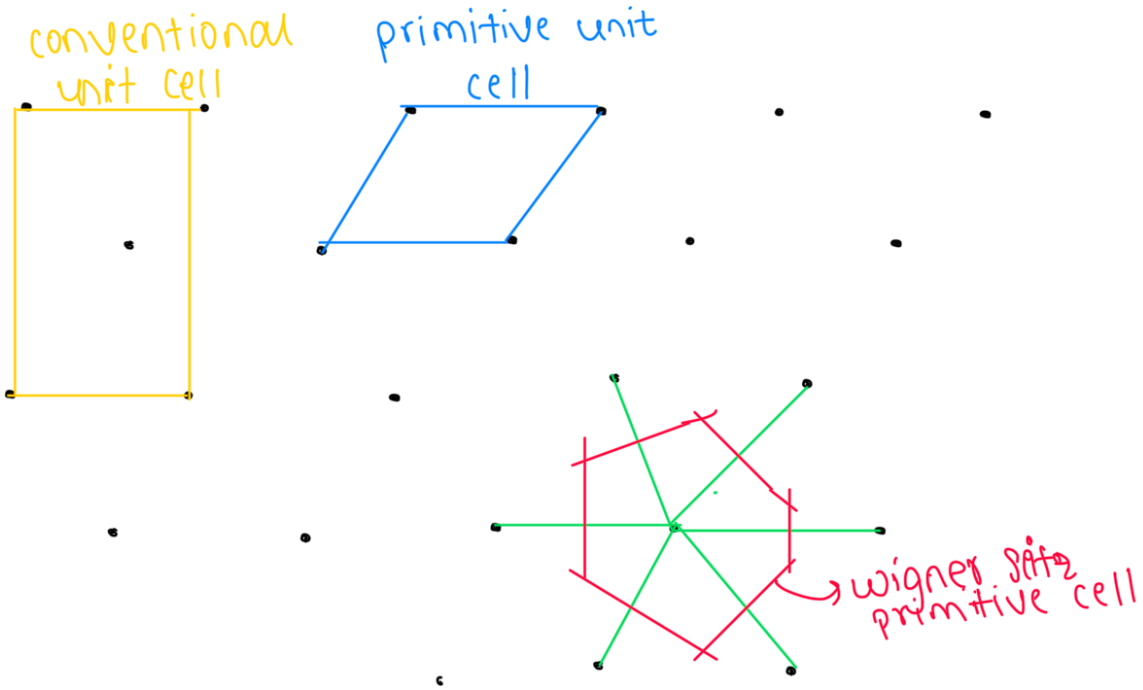
No of single particle states per unit energy is called density of states $g(\epsilon)$

Avg no of particles $N = \int g(\epsilon)\, \bar{n}_{FD}(\epsilon)\, d\epsilon$

**Lattice:** ∞ set of points defined by integer sums of a linearly independent primitive lattice vectors

**Unit Cell:** region of space such that when many identical units are stacked together, it tiles all of space

**Primitive Unit Cell:** unit cell containing exactly one lattice point.




For 3D ; wigner seitz unit cell of

→ BCC lattice ⇒ Truncated Octahedron

→ FCC lattice ⇒ Rhombic dodecahedron

Crystal Structure = lattice + Basis

## Lattice Plane and miller indices

find intercepts (x, y, z) along axes
divide by unit cell sizes : $x/a, y/b, z/c$
take reciprocals : $a/x, b/y, c/z$
reduce to smallest integer ratio h k l

angle b/w 2 directions

$$\cos\theta = \frac{h_1h_2 + k_1k_2 + l_1l_2}{(h_1^2 + k_1^2 + l_1^2)^{1/2}(h_2^2 + k_2^2 + l_2^2)^{1/2}}$$

all primitive cells have same density

## Reciprocal Lattice

Reciprocal domain (frequency) is used to represent time domain signals

$K = 2\pi/\lambda$

consider set of points R constituting a Bravais lattice and a plane wave $e^{ikr}$.

→ Generally K has no periodicity
→ for some wave vector G, $e^{iG\cdot R} = 1$ ; $e^{iG(r+R)} = e^{iGr}$

G is the set of all wave vectors that yield plane waves with the periodicity of a given

Bravais lattice

R → real space
G → reciprocal space

→ real space: $\vec{a_1}, \vec{a_2}, \vec{a_3}$

reciprocal space: $\vec{b_1}, \vec{b_2}, \vec{b_3}$

$$b_1 = \frac{2\pi(\vec{a_2} \times \vec{a_3})}{\vec{a_1} \cdot (\vec{a_2} \times \vec{a_3})}$$

$$b_2 = \frac{2\pi(\vec{a_3} \times \vec{a_1})}{\vec{a_2} \cdot (\vec{a_3} \times \vec{a_1})}$$

$$b_3 = \frac{2\pi(\vec{a_1} \times \vec{a_2})}{\vec{a_3} \cdot (\vec{a_1} \times \vec{a_2})}$$

$$a_i b_j = 2\pi \delta_{ij}$$